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# DEPENDENCE, EXTREMES, AND TEMPORAL CLUSTERING IN FRENCH AND GERMAN DAY-AHEAD ELECTRICITY PRICES IN 2025

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## ABSTRACT

This report studies hourly day-ahead electricity prices in France and Germany over the year 2025 using data from the ENTSO-E Transparency Platform. Our objective is to understand how market tensions appear, how they persist over time, and how they propagate across the two markets. To do so, we build a unified statistical analysis around a single dataset and a single research question: when the market becomes stressed, do French and German prices become more dependent, and do extreme episodes cluster in time?

We first clean and synchronize the hourly series, then study their temporal structure through standard linear time-series tools. We next analyze extreme price events using methods from extreme value theory in order to characterize rare positive spikes and negative prices, and to define meaningful thresholds for subsequent event-based modeling. We then investigate the cross-market dependence structure using copulas, comparing Gaussian, Gumbel, and Clayton specifications in both calm and stressed regimes. Finally, we model the temporal clustering of extreme events through univariate and bivariate Hawkes processes, which allow us to distinguish exogenous shocks from endogenous excitation and to quantify cross-market propagation.

Our results highlight three main features of the French-German day-ahead market in 2025: a strong common temporal structure, a dependence pattern that becomes especially relevant in the tails, and a clear clustering of extreme events over time. Together, these findings show that the interaction between the two markets cannot be reduced to simple linear correlation alone, but involves asymmetric tail dependence and self-exciting dynamics that are important for understanding stress episodes in the European electricity system.

## 1 Introduction

Electricity prices provide a natural setting for statistical analysis because they combine strong temporal regularities, abrupt stress episodes, and cross-market dependence. In European day-ahead markets, prices are determined on an hourly basis for the following day, which yields a regular time series that is well suited to linear time-series tools while also exhibiting extreme events such as very high prices or negative prices.

In this report, we focus on the French and German day-ahead electricity markets over the year 2025. Our goal is not to apply several disconnected methods to the same dataset, but rather to answer a single question: when market conditions become stressed, do the French and German price series become more dependent, and do extreme events propagate over time and across markets?

To address this question, we use a common cleaned dataset extracted from the ENTSO-E Transparency Platform and organize the analysis in four steps. We first study the marginal temporal structure of the hourly series using linear time-series methods. We then analyze the tails of the distribution using extreme-value tools. Next, we investigate cross-market dependence through copulas. Finally, we model the temporal clustering of extreme events with Hawkes processes. This progression reflects the logic of the course: margins before dependence, thresholds motivated by extreme-value analysis, and point-process modeling for temporal clustering that copulas alone cannot capture.

## 2 Data and Preprocessing

The data used in this report come from the ENTSO-E Transparency Platform<sup>1</sup>, under item `EnergyPrices_12.1.D_r3`. This dataset provides day-ahead electricity prices for each bidding zone and each market time unit, in currency per MWh. We downloaded the 12 monthly files corresponding to the year 2025 and concatenated them into a single panel dataset.

We then filtered the data to retain only the French and German bidding zones and restricted attention to hourly day-ahead observations. Timestamps were recorded in UTC and were carefully standardized before analysis. After merging the monthly files and removing possible duplicates by market zone and timestamp, we obtained two synchronized hourly series covering the full year 2025. Because of missing data points, the final sample size is 5855 observations for the German series and 6550 observations for the French series, which corresponds to a very high data quality given that there are 8760 hours in a year.

The final cleaned dataset serves as the common empirical basis for all parts of the report. This is important because the time-series, extreme-value, copula, and Hawkes analyses should not be viewed as separate projects, but as complementary perspectives on the same underlying market dynamics.

## 3 Linear Time-Series Analysis

### 3.1 Framework

In order to discuss the dynamics of the data based on time-series analysis, let us first explain some notations. Whether it is for french or german energy prices, we'll consider the time space:

$$\mathcal{T} = \{\text{hour} : \text{min} \mid \text{hour} \in \llbracket 0, 23 \rrbracket, \text{min} \in \{0, 15, 30, 45\}\}.$$

Every process  $X$  considered in this section will take as events the times in the space  $\mathcal{T}$ , and will span from day 1 to day 365, meaning we'll write  $(X_t)_t$  instead of  $(X_t)_{t \in \llbracket 1, 365 \rrbracket}$ . Moreover, expectancies will be taken as if the times were chosen uniformly at random in  $\mathcal{T}$ .

### 3.2 Test of stationarity

#### 3.2.1 Expectancy function

In order to modelise our data as  $\text{ARMA}(p, q)$  processes, we first need to assess if we can assume the data to be stationary. Let's recall that a process  $(X_t)_t$  is said to be (weakly-)stationary if it's square-integrable, its expectancy function is constant and for all  $0 \leq s \leq t$ ,  $\text{Cov}(X_s, X_t)$  only depends on  $t - s$ . To test the last two hypothesis, we need to run the computation of the expectancies and the autocovariance function.

Table 1: Means and standard deviation of the mean prices per day.

| Country            | Mean price (€) | Ratio std/mean |
|--------------------|----------------|----------------|
| France             | 61.08          | 0.32           |
| Germany-Luxembourg | 89.39          | 0.31           |

<sup>1</sup>[https://transparency.entsoe.eu/fileLibrary/fileBrowserMode?appState=%7B%22path%22%3A%22%2FFTP\\_export%2FEnergyPrices\\_12.1.D\\_r3%2F%22%7D](https://transparency.entsoe.eu/fileLibrary/fileBrowserMode?appState=%7B%22path%22%3A%22%2FFTP_export%2FEnergyPrices_12.1.D_r3%2F%22%7D)

The ratio between standard deviation and the mean helps on showing if the function  $t \mapsto \mathbb{E}(X_t)$  is constant or not. Here, the ratio might seem too high. To try to circumvent this issue, we introduce the differentiation operator

$$\nabla: X \mapsto (X_t - X_{t-1})_t.$$

By applying this operator  $d \in \mathbb{N}^*$  times to a process  $X$ , it removes polynomial tendencies of degree  $d - 1$ . Then, for example the process  $\nabla^2 X$  could prove to have its expectancy function constant, and we could then try to modelise it as an ARMA process. To retrieve the initial process, one may note that:

$$Y_t = Y_0 + \sum_{i=1}^t (\nabla Y)_i.$$

Figure 1 shows the ratio between standard deviation and mean on the function  $t \mapsto \mathbb{E}((\nabla^d X)_t)$  for  $d$  varying from 0 to 100, while figure 2 shows the same for Germany. One may notice that the two best ratios are obtained for  $d = 0$  and  $d = 1$ , meaning we'll be modelising one of the two as an ARMA process. The next step is to compute the autocovariance function.

### 3.2.2 Autocovariance function

For a stationary process, computing the autocovariance function is fairly easy. One only needs to compute:

$$\gamma(n) = \text{Cov}(X_n, X_0), \quad \text{for every } n \in \mathbb{N}.$$

However, to test the stationarity, we'll be computing for every  $n \in \mathbb{N}$ ,

$$t \mapsto \text{Cov}(X_{t+n}, X_t),$$

to test whether it is constant or not. We did this computation, and checked the mean and the standard deviation, to figure out which trajectory would be best. Figure 3 shows the ratio between standard deviation and mean for the autocovariance function, while figure 4 shows the same for Germany. The trajectories depend on the number of times the process has been differentiated. It clearly shows that the best candidate to be modelised as an ARMA is the energy prices process itself.

## 3.3 ARMA modelisation

### 3.3.1 With statsmodels

The module `statsmodels` is built in Python and provides classes and functions for the estimation of many different statistical models, as well as for conducting statistical tests, and statistical data exploration. In particular, it comes with the extremely useful `tsa.arma` function which computes the best fitting ARMA model, given values of  $p$  and  $q$ , minimizing the Akaike information criterion (AIC). This criterion is computed according to this formula:

$$\text{AIC} = 2N - 2\ln(L),$$

where  $N$  is the number of estimated parameters and  $L$  is the maximum of the likelihood function. Given the computation complexity of this function, we ran it over the tuples in  $\llbracket 0, 5 \rrbracket^2$  to get the best fitting model. Results can be seen in figure 5 for the french prices, and figure 6 for german prices.

### 3.3.2 With an AR(p) process

An  $\text{AR}(p)$  process is an  $\text{ARMA}(p, q)$  process in which we let  $q = 0$ . In this subsection, we try to find an AR model which could perform better than the  $\text{ARMA}(5, 5)$  previously found with the `statsmodels` module. Suppose we have a process  $X$  being  $\text{AR}(p)$  with  $p \in \mathbb{N}$ . Then we can write

$$X_t = \sum_{i=1}^p \phi_i X_{t-i} + \epsilon_t,$$

where  $\epsilon$  is a white noise of standard deviation  $\sigma$ . We denote by  $\gamma$  the autocovariance function. Then, for all  $m \in \mathbb{N}$ ,

$$\begin{aligned} \mathbb{E}(X_t X_{t-m}) &= \sum_{i=1}^p \phi_i \mathbb{E}(X_{t-m} X_{t-i}) + \mathbb{E}(\epsilon_t X_{t-m}) \\ \gamma_m &= \sum_{i=1}^p \phi_i \gamma_{m-i} + \sigma^2 \delta_{m,0}, \end{aligned}$$

by definition of  $\gamma$  and using the result in the course for the second term. These equations are known as Yule-Walker equations. Now, let us consider the sequence of matrices defined by

$$\Gamma_m = \begin{pmatrix} \gamma_0 & \gamma_{-1} & \cdots & \gamma_{1-m} \\ \gamma_1 & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \gamma_{m-1} & \cdots & \cdots & \gamma_0 \end{pmatrix}, \quad \text{for all } m \in \mathbb{N}.$$

Then, one gets from the Yule-Walker equations that for every  $m \in \mathbb{N}^*$ ,

$$(\gamma_1 \quad \cdots \quad \gamma_m)^T = \Gamma_m (\phi_1 \quad \cdots \quad \phi_m)^T.$$

Assuming  $\Gamma_m$  is invertible:

$$(\phi_1 \quad \cdots \quad \phi_m)^T = \Gamma_m^{-1} (\gamma_1 \quad \cdots \quad \gamma_m)^T.$$

Then we get, for  $m = 0$ ,

$$\sigma^2 = \gamma_0 - \sum_{i=1}^p \phi_i \gamma_{-i}.$$

We can use that  $\gamma_n = \gamma_{-n}$  for every  $n \in \mathbb{Z}$ . Then we compute the AIC for each model  $p$ , using that we estimated  $N = p + 1$  parameters, i.e.:

$$\text{AIC}_p = 2(p + 1) + (T - p) \ln(\sigma^2) + (T - p)(1 + \ln(2\pi)),$$

here  $T = 365$  (number of days in a year, sample size), and we subtract  $p$  because the first  $p$  observations are fixed. Figure 7 shows the results for french prices, while figure 8 shows the results for german prices. It is clear that the AR model is better with increasing  $p$ . However, we cannot compare the values of AIC between this computation and the one of `statsmodels` module, because they may differ from a constant.

To check whether our results are compatible with `statsmodels`, we ran the code over the best fitting AR model for  $p$  varying from 0 to 25. Figure 9 shows the results for french prices, while figure 10 show the results for german prices. In both cases, the best model is AR(25), which is compatible with the two graphs obtained with the Yule-Walker equations. Secondly, it's interesting to note that:

$$\text{AIC}(\text{AR}(25)) < \text{AIC}(\text{ARMA}(5, 5))$$

for french prices, as well as german prices. However, even though we ran the tests over 25 values of  $p$ , instead of 36 values of  $(p, q)$ , the code took 50% more time to compile. Thus, it should be best to use ARMA modelisation rather than AR, especially for large values of  $p$ , and one can use Yule-Walker equations to get a faster result.

### 3.4 GARCH modelisation

As the data processed consists of prices, one may argue that ARMA modelisation is not adapted, and that we need to modelise it by a GARCH model. This section quickly explores this case. We used the library `arch` to implement this.

Figures 11 and 12 show the fitting of a GARCH(1, 1) model on the french prices, with the prices on each day being averaged. Then, the model predicts what could happen in the next month. Seeing the graphs, it seems that modelling the prices as a GARCH is much more interesting than an ARMA, as our data isn't clearly stationary in the first case. In addition, finding a linear invertible transform that can make the data stationary is hard. In GARCH case, the fitting is believable, even though that's not nearly the case for the forecasting.

Figures 13 and 14 show the fitting of a GARCH(2, 2) model on the french prices ; while figures 15 and 16 show the fitting of a GARCH(10, 10) model on the french prices. We see that both models GARCH(2, 2) and GARCH(10, 10) act nearly similarly on the french prices. However, the forecastings differ completely. On one hand, GARCH(2, 2) predicts a drop in the prices which seems linear, contrary to the prediction of a quadratic drop with GARCH(1, 1). On the other hand, GARCH(10, 10) seems to be predicting the average of the time-series we analysed. As the processes act as white noises, it's not surprising to see the forecast converging to a constant value.

Thus, in our case, the french and german energy prices might be better off modelised by a GARCH model, which is in agreement with the course we took: financial time series are not compatible with ARMA processes, that's why GARCH processes were invented in the first place.

The linear time-series analysis provides the first description of the marginal dynamics of the two series. However, average dynamics are not sufficient to characterize market stress episodes, which motivates a specific focus on extreme observations.

## 4 Extreme-Value Analysis

Once the marginal temporal structure of the two price series has been established, the natural next step is to move beyond average dynamics and focus on the tails of the distribution. In this section we therefore study the distributional behaviour of extreme observations in both the upper tail (very high prices) and the lower tail (mainly negative prices) using the Peaks-over-Threshold (POT) framework from Extreme Value Theory (EVT).

The section is organised as follows. We first recall the theoretical foundations of the POT method and the Generalised Pareto Distribution (GPD). We then carry out the threshold selection, the GPD estimation and the goodness-of-fit diagnostics separately for the upper and lower tails of the French and German series. We complement the maximum-likelihood estimates with the semi-parametric Hill estimator. Finally, we draw the implications of this analysis for the subsequent copula and Hawkes sections.

### 4.1 The Peaks-over-Threshold Framework

The Peaks-over-Threshold method rests on the **Pickands–Balkema–de Haan theorem**, which states that, for a sufficiently high threshold  $u$ , the conditional distribution of the exceedances

$$Y = X - u \quad \text{for } X > u \quad (1)$$

converges to a **Generalised Pareto Distribution (GPD)**:

$$F_{\xi, \beta}(y) = 1 - \left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi}, \quad y > 0, \quad \beta > 0, \quad (2)$$

where  $\xi \in \mathbb{R}$  is the *shape* parameter and  $\beta > 0$  is the *scale* parameter. The sign of  $\xi$  determines the tail regime:  $\xi > 0$  corresponds to a heavy-tailed (Fréchet) distribution with a polynomial decay;  $\xi = 0$  corresponds to an exponential tail (Gumbel); and  $\xi < 0$  corresponds to a bounded upper tail (Weibull).

A key diagnostic tool for threshold selection is the **Mean Excess Plot (MEP)**, which traces the empirical mean excess function  $e(u') = E[X - u' \mid X > u']$  as a function of the threshold  $u'$ . Under the GPD model,  $e(u')$  is approximately linear for sufficiently high thresholds. In practice, we select a threshold in the region where the MEP exhibits approximate linearity: this corresponds to a value of  $u'$  large enough that the GPD provides a reasonable approximation, while retaining enough exceedances for reliable estimation.

**Lower-tail exceedances.** The same framework applies to the lower tail via the standard reflection trick. For a low threshold  $u$ , the *lower exceedances* (or deficits) are defined as

$$Y = u - X \quad \text{for } X < u, \quad (3)$$

which is equivalent to applying the POT method to the mirrored series  $-X$ .

**Maximum-likelihood estimation.** Given a threshold  $u$  and the resulting sample of exceedances  $y_1, \dots, y_{n_u}$ , we estimate  $(\xi, \beta)$  by maximising the log-likelihood

$$\ell(\xi, \beta) = -n_u \log \beta - \left(1 + \frac{1}{\xi}\right) \sum_{i=1}^{n_u} \log \left(1 + \frac{\xi y_i}{\beta}\right), \quad (4)$$

subject to  $\beta > 0$  and  $1 + \xi y_i/\beta > 0$  for all  $i$ . The optimisation is initialised from multiple starting points to guard against local optima.

### 4.2 Upper Tail: Very High Prices

**Threshold selection.** Figure 17 shows the Mean Excess Plot for the French and German series. The plot displays the empirical mean excess function over a grid of quantile thresholds ranging from the 75th to the 99.5th percentile. It is not obvious to choose a suitable threshold because the curve does not exhibit a single clear linear region.

To address this issue, we compute the GPD parameter estimates over the same grid of quantile thresholds. Figures 18 and 19 display the estimated parameters as a function of the quantile threshold  $q$ , for  $q \in [0.80, 0.99]$ . A well-chosen threshold corresponds to a high enough quantile which lies in a region of relative stability in both parameter estimates before the variance explodes as fewer exceedances remain. Based on this stability analysis, we select  $q = 0.93$  for both France and Germany, which yields the thresholds reported in Table 2.

Table 2: Selected thresholds and number of upper-tail exceedances at quantile  $q = 0.93$ .

| Market  | Threshold $u$ (EUR/MWh)       | Number of exceedances $n_u$ |
|---------|-------------------------------|-----------------------------|
| France  | $u_{FR} = 133.52$ (93rd pct.) | 500                         |
| Germany | $u_{DE} = 151.22$ (93rd pct.) | 410                         |

**GPD fit.** Figure 20 shows the histogram of the exceedances together with the fitted GPD density for France and Germany. The GPD provides a good description of the empirical distribution of upper-tail exceedances in both markets. Figure 21 presents the corresponding QQ-plots, which confirm that the fitted model aligns well with the empirical quantiles, except for the very largest observations where the noise is inherently large given the small sample size at the extreme end.

### 4.3 Lower Tail: Negative Prices

**Threshold selection.** The lower-tail analysis mirrors the upper-tail procedure. Figure 22 displays the lower-tail analogue of the Mean Excess Plot over a grid of quantile thresholds  $q \in [0.005, 0.25]$ . The approximately linear region lies clearly to the right of zero in both panels, which points to close to  $u = 0$  as the natural threshold for the lower-tail analysis.

Figures 23 and 24 show the estimated shape and scale parameters as a function of the lower quantile threshold. Stability is reached around  $q = 0.025$  for both France and Germany. This threshold is selected for the final estimation and is reported in Table 3.

Table 3: Selected thresholds and number of lower-tail exceedances at quantile  $q = 0.025$ .

| Market  | Threshold $u$ (EUR/MWh)       | Number of exceedances $n_u$ |
|---------|-------------------------------|-----------------------------|
| France  | $u_{FR} = -1.67$ (2.5th pct.) | 169                         |
| Germany | $u_{DE} = -6.60$ (2.5th pct.) | 147                         |

**GPD fit.** Figure 25 shows the histogram of lower-tail deficits and the fitted GPD density for both markets. Figure 26 provides the corresponding QQ-plots. As with the upper tail, the fit is satisfactory across most of the distribution, with the usual instability at the most extreme quantiles.

### 4.4 Robustness: The Hill Estimator

As an additional check on the GPD maximum-likelihood estimates, we compute the **Hill estimator**, a semi-parametric estimator of the tail index valid in the heavy-tailed (Fréchet) domain  $\xi > 0$ . For a sample  $X_{(1)} \geq X_{(2)} \geq \dots \geq X_{(n)}$  of order statistics, the Hill estimator with  $k$  upper-order statistics is

$$\hat{\xi}_H(k) = \frac{1}{k} \sum_{i=1}^k \log X_{(i)} - \log X_{(k+1)}. \quad (5)$$

The Hill plot traces  $\hat{\xi}_H(k)$  as a function of  $k$  and the optimal estimate is read from the stable plateau of this curve. For the lower tail, the estimator is applied to the reflected series  $-X$ . Figures 27 and 28 display the Hill plots for the upper and lower tails respectively.

### 4.5 Summary of Tail Parameter Estimates

Table 4 consolidates the shape parameter estimates from both methods for each market and each tail.

Table 4: Estimated tail shape parameters  $\hat{\xi}$  from GPD maximum likelihood and from the Hill estimator, for the upper and lower tails of the French and German day-ahead price series.

| Market  | Tail  | $\hat{\xi}$ (GPD MLE) | $\hat{\xi}$ (Hill) |
|---------|-------|-----------------------|--------------------|
| France  | Upper | 0.0735                | 0.1514             |
| Germany | Upper | 0.3601                | 0.2621             |
| France  | Lower | 0.7185                | 1.2708             |
| Germany | Lower | 0.3842                | 0.8833             |

#### 4.6 Interpretation

Several conclusions emerge from this analysis.

**All tails are heavy.** Every estimated shape parameter satisfies  $\hat{\xi} > 0$ , placing both markets unambiguously in the Fréchet domain of attraction. This confirms that day-ahead electricity prices exhibit power-law behaviour in both tails, which has direct consequences for risk management and market design.

**Strong asymmetry between upper and lower tails.** A structural asymmetry emerges between the two directions of extreme events. For both France and Germany, the lower tail is substantially heavier than the upper tail:

$$\hat{\xi}_{\text{lower}} \gg \hat{\xi}_{\text{upper}}.$$

This finding is meaningful. Negative price episodes appear to generate more extreme deviations relative to their threshold than positive price spikes. In other words, when prices fall below zero, they can fall very far; the lower tail exhibits more pronounced extremity than the upper tail.

**Structural differences between France and Germany.** Beyond the common heavy-tail finding, the two markets differ in their relative tail behaviour. For the upper tail, Germany exhibits a heavier tail than France:

$$\hat{\xi}_{\text{DE}}^{\text{upper}} > \hat{\xi}_{\text{FR}}^{\text{upper}},$$

indicating that, conditional on exceeding a selected threshold, very large positive price spikes tend to be more extreme in the German market. Conversely, for the lower tail, France appears to exhibit heavier extremes:

$$\hat{\xi}_{\text{FR}}^{\text{lower}} > \hat{\xi}_{\text{DE}}^{\text{lower}},$$

suggesting that extreme negative price episodes are relatively more pronounced in France once they materialise. These cross-country asymmetries motivate the regime-conditioned copula analysis of Section 5.

**Robustness of the EVT characterisation.** Despite methodological differences (the Hill estimator focuses exclusively on the most extreme observations, whereas the GPD fit incorporates a broader set of threshold exceedances) both approaches deliver qualitatively consistent conclusions. They agree on the presence of heavy tails, on the asymmetry between upper and lower tails, and on the relative ordering of tail heaviness across the two markets. The Hill estimates tend to be somewhat larger, as expected from the fact that this estimator is sensitive to the very outermost observations and can be upward-biased in finite samples. The GPD MLE is therefore our preferred point estimate, but the Hill comparison serves as a useful robustness check.

**Implications for the subsequent analyses.** The EVT analysis characterises the marginal tail behaviour of each series independently. It does not address whether extreme events occur jointly across markets or whether they cluster in time. However, the presence of heavy tails has two direct implications for the rest of the report. First, since linear correlation is blind to the dependence structure in the tails, the heavy-tail finding strengthens the case for using copulas to study cross-market dependence. Second, since the tail shape parameters confirm that price spikes are not isolated outliers but rather draw from heavy-tailed distributions, the temporal clustering of these events, studied through Hawkes processes in Section 6, is a natural follow-up question.

## 5 Copula-Based Dependence Analysis

We now turn to the dependence structure between the French and German day-ahead electricity markets. After studying the marginal temporal dynamics and the distributional behavior of extreme observations, our goal is to characterize the joint dependence between the two markets in a way that goes beyond linear correlation.

The copula framework is particularly well suited to this task because it separates marginal behavior from the dependence structure. In our setting, this allows us to study whether French and German prices become jointly extreme during stress episodes, and whether this dependence is symmetric or concentrated in a specific tail.

### 5.1 Variables and pseudo-observations

To reduce the impact of deterministic seasonal effects, we work with de-seasonalized hourly price variations for France and Germany, denoted by  $X_t$  and  $Y_t$ . These transformed variables are intended to capture the short-run residual market fluctuations after removing regular hourly and weekly patterns.

Following the copula methodology, we construct pseudo-observations

$$\hat{U}_t = \hat{F}_X(X_t), \quad \hat{V}_t = \hat{F}_Y(Y_t),$$

where  $\hat{F}_X$  and  $\hat{F}_Y$  are the empirical marginal distribution functions. This step maps the original observations to the unit square and isolates the dependence structure from the marginal scales.

Figure 29 displays both the scatter plot of the transformed French and German variations and the corresponding pseudo-observations. The raw scatter plot already suggests a positive dependence, while the pseudo-observation plot reveals that the dependence is not purely elliptical and may be stronger in the upper tail.

### 5.2 Rank-based dependence measures

We begin with nonparametric rank-based indicators. On the full sample, the Kendall coefficient is estimated at

$$\hat{\tau} = 0.2177,$$

while the Spearman coefficient is

$$\hat{\rho} = 0.3062.$$

These values indicate a positive but moderate dependence between the two markets. The two price series are clearly linked, but not in a way that would suggest near-deterministic co-movement. This makes the copula approach relevant: there is enough dependence to be modeled, but enough flexibility left to distinguish between alternative dependence structures.

### 5.3 Parametric copula families

We estimate three parametric copula families that were studied in the course: the Gaussian copula, the Gumbel copula, and the Clayton copula. The Gaussian copula is a natural benchmark but does not exhibit tail dependence. By contrast, the Gumbel copula captures upper-tail dependence, whereas the Clayton copula captures lower-tail dependence. This distinction is especially relevant for electricity prices, where the economic interpretation of very high prices and very low or negative prices is quite different.

For each family, we consider two estimation strategies: inversion of Kendall's tau and maximum likelihood based on the pseudo-observations. We then compare the fitted models through their log-likelihood, and implied tail-dependence coefficients.

Table 5 reports the results on the full sample.

Table 5: Copula estimation results on the full sample.

| Family   | $\hat{\tau}$ | $\hat{\rho}$ | Param. (tau) | Param. (MLE) | $\ell_{\text{tau}}$ | $\ell_{\text{MLE}}$ | $\lambda_l$ | $\lambda_u$ |
|----------|--------------|--------------|--------------|--------------|---------------------|---------------------|-------------|-------------|
| Gaussian | 0.2177       | 0.3062       | 0.3353       | 0.3113       | 294.6647            | 297.0064            | 0.0000      | 0.0000      |
| Gumbel   | 0.2177       | 0.3062       | 1.2783       | 1.2773       | 392.1663            | 392.1692            | 0.0000      | 0.2794      |
| Clayton  | 0.2177       | 0.3062       | 0.5566       | 0.3938       | 215.6555            | 245.2915            | 0.1720      | 0.0000      |



The ranking is unambiguous on the full sample: the Gumbel copula clearly dominates the Gaussian and Clayton alternatives. It yields the highest log-likelihood, which indicates that the dependence between the two markets is better described by an asymmetric structure with upper-tail dependence than by a symmetric Gaussian benchmark or a lower-tail-dependent Clayton copula.

This result is economically meaningful. It suggests that the strongest common dependence between the French and German markets does not occur primarily during jointly very low prices, but rather during jointly high price episodes. In other words, the two markets appear especially connected when upward stress affects both systems at the same time.

#### 5.4 Calm versus stressed regimes

To go beyond the average dependence structure, we split the sample into a calm regime and a stressed regime, based on large spread or volatility conditions. The purpose is not only to determine which copula performs best overall, but also to understand whether the form of dependence changes across market regimes.

Table 6 summarizes the corresponding estimates.

Table 6: Copula estimation results by regime.

| Regime   | Family   | $\hat{\tau}$ | $\hat{\rho}$ | Param. (MLE) | $\ell_{\text{MLE}}$ | $\lambda_l$ | $\lambda_u$ |
|----------|----------|--------------|--------------|--------------|---------------------|-------------|-------------|
| Calm     | Gaussian | 0.2231       | 0.3168       | 0.3659       | 237.2648            | 0.0000      | 0.0000      |
| Calm     | Gumbel   | 0.2231       | 0.3168       | 1.3341       | 299.8596            | 0.0000      | 0.3187      |
| Calm     | Clayton  | 0.2231       | 0.3168       | 0.5021       | 195.2759            | 0.2514      | 0.0000      |
| Stressed | Gaussian | 0.2073       | 0.2888       | 0.2364       | 75.9999             | 0.0000      | 0.0000      |
| Stressed | Gumbel   | 0.2073       | 0.2888       | 1.1926       | 107.9107            | 0.0000      | 0.2118      |
| Stressed | Clayton  | 0.2073       | 0.2888       | 0.2633       | 67.7384             | 0.0719      | 0.0000      |

Several points emerge from this comparison. First, the Gumbel copula remains the best-performing family in both regimes, which makes the main conclusion robust. Second, the overall rank dependence is slightly stronger in the calm regime than in the stressed regime, as seen from both Kendall's tau and Spearman's rho. Third, upper-tail dependence remains present in both regimes, but is estimated to be larger in the calm regime than in the stressed one.

At first sight, the last point may seem surprising. One might expect stronger upper-tail dependence precisely during stressed periods. However, our stress partition captures high-volatility or large-spread situations more broadly, which may include heterogeneous episodes in which the two markets do not always move to the same extreme at the same time. In that sense, the stressed regime may contain more dispersion and more local asymmetries, even though it remains more turbulent.

#### 5.5 Empirical copula diagnostics and tail concentration

Figure 30 shows the empirical copula heatmap, while Figure 31 presents the fitted contours for the three parametric families. These figures are useful because they complement the likelihood-based comparison with a direct visual representation of the dependence structure.

We also examine empirical tail concentration functions on Figure 32 in order to compare the lower and upper corners of the unit square. The lower-tail concentration decreases slowly and then appears to converge to zero for very small values of  $q$ , which is not consistent with a very strong lower-tail dependence. By contrast, the upper-tail concentration is more structured: on the full sample it remains clearly positive, while in the stressed regime it becomes more irregular and even increases at very small values of  $q$ . Although the smallest- $q$  region must be interpreted cautiously because of limited sample size, this pattern is coherent with the dominance of the Gumbel family.

#### 5.6 Interpretation

Overall, the copula analysis shows that the dependence between French and German day-ahead prices cannot be reduced to average linear co-movement. The best-fitting copula is consistently the Gumbel copula, which points to an asymmetric dependence structure with upper-tail dependence. In economic terms, this means that the most informative joint behavior appears during common upward price stress, rather than during jointly very low or negative prices.

This conclusion complements the previous sections of the report. Once the marginal time-series dynamics have been accounted for, and once rare events have been identified through the extreme-value analysis, the copula framework

reveals that dependence itself becomes more informative in the upper tail. This motivates the final step of our study: if extreme events occur jointly across markets, do they also cluster in time? This is the question addressed by the Hawkes-process analysis.

## 6 Hawkes Processes for Extreme Event Clustering

Copulas describe cross-market dependence, but they do not capture the temporal clustering of extreme events. In electricity markets, this temporal dimension is essential: once a market enters a stressed state, extreme prices often do not occur in isolation, but rather in bursts. For this reason, we model extreme events using Hawkes processes.

A Hawkes process is a self-exciting point process whose intensity depends on past events. In the multivariate setting, each component may also be excited by events occurring in the other component, which makes the framework particularly suitable for studying cross-market propagation. The process is stationary when the spectral radius of the branching matrix is strictly less than one.

### 6.1 From hourly prices to event times

Rather than modeling prices themselves as a point process, we transform the hourly time series into timestamped events. This is more natural in the Hawkes framework and better aligned with the objective of studying clustering in rare episodes.

We consider several event definitions, including negative prices, very high prices above the empirical 99% quantile, large hourly jumps, and extreme spreads. In the final analysis, we focus on two economically meaningful cases: negative prices and very high prices. The negative-price case captures episodes of excess supply or severe market imbalance, while the high-price case captures upward stress episodes.

Table 7 summarizes the frequency of the different event types.

Table 7: Number and share of extreme events in France and Germany.

| Event type         | FR count | DE count | FR share | DE share |
|--------------------|----------|----------|----------|----------|
| Negative price     | 434      | 459      | 0.0741   | 0.0784   |
| High price 95%     | 293      | 293      | 0.0500   | 0.0500   |
| Jump 95%           | 293      | 293      | 0.0500   | 0.0500   |
| Spread extreme 95% | 293      | 293      | 0.0500   | 0.0500   |
| High price 99%     | 59       | 59       | 0.0101   | 0.0101   |

These frequencies justify the event-based approach. Negative prices are rare enough to be interpreted as stress events, but still numerous enough to support estimation. The same is true, more restrictively, for the upper 1% price events.

### 6.2 Univariate Hawkes analysis: negative prices

We begin with univariate Hawkes processes estimated separately on the French and German negative-price events. We use an exponential kernel

$$\phi(t) = \alpha e^{-\beta t} \mathbf{1}_{t>0},$$

so that the branching ratio is given by  $\alpha/\beta$ .

The observation window spans 6549 hours, with 434 negative-price events in France and 459 in Germany. Table 8 reports the corresponding estimates.

Table 8: Univariate Hawkes estimates for negative-price events.

| Country | $\mu$  | $\alpha$ | $\beta$ | Branching ratio | $\bar{\lambda}$ | Endogenous share | Log-likelihood |
|---------|--------|----------|---------|-----------------|-----------------|------------------|----------------|
| France  | 0.0139 | 0.4935   | 0.6240  | 0.7909          | 0.0663          | 0.7909           | -1073.1962     |
| Germany | 0.0148 | 0.5065   | 0.6415  | 0.7895          | 0.0701          | 0.7895           | -1119.8032     |

The baseline intensities are small in both countries, whereas the branching ratios are close to 0.79. This is a strong indication of self-excitation: once a negative-price event occurs, the probability of observing another similar event shortly afterward increases substantially. In both markets, roughly four-fifths of the total event activity is attributed to endogenous dynamics rather than to exogenous shocks.

This result confirms that negative prices are temporally clustered. They do not arrive as isolated points in time, but rather in bursts, which is exactly the kind of structure that a Hawkes process is designed to detect.

Figure 33 presents the event raster plot over the full sample, while Figure 34 shows the fitted univariate intensities.

### 6.3 Bivariate Hawkes analysis: negative prices

We then estimate a bivariate Hawkes process in order to distinguish self-excitation from cross-excitation between the French and German markets. The fitted model yields the following branching matrix:

$$G = \begin{pmatrix} 0.6886 & 0.1321 \\ 0.0308 & 0.7672 \end{pmatrix},$$

with spectral radius

$$\rho(G) = 0.8028 < 1.$$

The stationarity condition is therefore satisfied. The model is stable, and the estimated long-run average intensities are  $\bar{\lambda}_{FR} \approx 0.0663$  and  $\bar{\lambda}_{DE} \approx 0.0701$ .

Several features are noteworthy. First, both diagonal terms are large, which confirms strong self-excitation in each market. Second, cross-excitation is clearly asymmetric: the effect from Germany to France is larger than the reverse effect. More precisely, an extreme negative-price event in Germany appears to increase the short-term probability of a corresponding event in France more strongly than the opposite channel.

This asymmetry is economically plausible. It suggests that, for negative prices, the German market plays a stronger propagating role within the French-German pair. At the same time, the diagonal terms remain dominant, which means that local persistence is stronger than cross-border contagion.

Table 9 summarizes the bivariate estimates.

Table 9: Bivariate Hawkes estimates for negative-price events.

| Country | $\mu$  | $\bar{\lambda}$ | Endogenous share |
|---------|--------|-----------------|------------------|
| France  | 0.0114 | 0.0663          | 0.8283           |
| Germany | 0.0143 | 0.0701          | 0.7963           |

Table 10: Estimated excitation matrix for negative-price events.

| Target / Source | FR source | DE source |
|-----------------|-----------|-----------|
| FR target       | 0.6886    | 0.1321    |
| DE target       | 0.0308    | 0.7672    |

Figure 35 visualizes this excitation matrix, and Figure 36 provides a zoom on the most tense week of the year.

### 6.4 Very high prices: upper 1% events

We perform the same analysis on very high prices, defined as observations above the empirical 99% quantile. In this case, the observation window is again 6549 hours, but only 59 events are recorded in each country. These events are therefore much rarer than negative prices, which makes them especially relevant as indicators of severe upward market stress.

The univariate estimates are reported in Table 11. They again reveal self-excitation, with branching ratios of 0.8391 for France and 0.6663 for Germany.

Table 11: Univariate Hawkes estimates for high-price 99% events.

| Country | $\mu$  | $\alpha$ | $\beta$ | Branching ratio | $\bar{\lambda}$ | Endogenous share | Log-likelihood |
|---------|--------|----------|---------|-----------------|-----------------|------------------|----------------|
| France  | 0.0014 | 0.1001   | 0.1193  | 0.8391          | 0.0090          | 0.8391           | -221.1900      |
| Germany | 0.0031 | 0.1242   | 0.1864  | 0.6663          | 0.0093          | 0.6663           | -250.7652      |

The French series exhibits particularly strong self-excitation for upper-tail events. Once a very high price occurs in France, the fitted model indicates a high short-run probability of another high-price event shortly afterward. Germany also displays clustering, though less strongly.

The bivariate model leads to the branching matrix

$$G = \begin{pmatrix} 0.8271 & 0.0128 \\ 0.1739 & 0.5234 \end{pmatrix},$$

with spectral radius

$$\rho(G) = 0.8343 < 1.$$

Again, the stationarity condition is satisfied. The diagonal terms are large, especially for France, which confirms strong within-market clustering of upper-tail events. The cross-excitation terms are smaller than in the negative-price case and are also asymmetric, but now in the opposite direction: the effect from France to Germany is larger than the effect from Germany to France.

This difference between negative-price events and upper-tail events is particularly interesting. It suggests that the direction of cross-market propagation depends on the nature of the stress event. Negative-price episodes appear more strongly transmitted from Germany toward France, whereas very high price events appear more strongly transmitted from France toward Germany.

The corresponding summary tables are reported below.

Table 12: Bivariate Hawkes estimates for high-price 99% events.

| Country | $\mu$  | $\bar{\lambda}$ | Endogenous share |
|---------|--------|-----------------|------------------|
| France  | 0.0014 | 0.0090          | 0.8401           |
| Germany | 0.0028 | 0.0091          | 0.6957           |

Table 13: Estimated excitation matrix for high-price 99% events.

| Target / Source | FR source | DE source |
|-----------------|-----------|-----------|
| FR target       | 0.8271    | 0.0128    |
| DE target       | 0.1739    | 0.5234    |

## 6.5 Interpretation

Taken together, the Hawkes analysis shows that extreme events are strongly clustered in time and exhibit measurable cross-excitation between the French and German markets. This cross-market propagation is asymmetric and depends on the event definition: negative-price episodes appear to spill over more strongly from Germany to France, whereas upper-tail events show the opposite pattern. In this sense, the Hawkes framework complements the copula analysis by adding a temporal dimension to joint extremes, showing that stress in the French-German day-ahead market is not only cross-sectional but also dynamically propagated over time.

## 7 Results and Discussion

The four methods used in this report provide complementary information on the same empirical question. The linear time-series analysis highlights the strong hourly and weekly structure of the market. The extreme-value analysis shows that rare and economically meaningful stress episodes remain an essential feature of the 2025 price distribution. The

copula analysis reveals that dependence between France and Germany is not fully summarized by linear correlation and becomes especially informative in the tails. Finally, the Hawkes analysis shows that extreme events are temporally clustered and partially transmitted across markets.

Taken together, these findings suggest that market stress in the French-German electricity system has both a cross-sectional and a temporal dimension. Dependence matters because prices can become jointly extreme, and self-excitation matters because extreme events tend to arrive in bursts rather than in isolation. This joint perspective is one of the main contributions of the report.

## Data and Code Availability

The data used in this report come from the ENTSO-E Transparency Platform, item `EnergyPrices_12.1.D_r3`, which provides day-ahead electricity prices by bidding zone and market time unit. The monthly files for 2025 can be accessed through the ENTSO-E file library at:

[https://transparency.entsoe.eu/fileLibrary/fileBrowserMode?appState=%7B%22path%22%3A%22%2FFTP\\_export%2FEnergyPrices\\_12.1.D\\_r3%2F%22%7D](https://transparency.entsoe.eu/fileLibrary/fileBrowserMode?appState=%7B%22path%22%3A%22%2FFTP_export%2FEnergyPrices_12.1.D_r3%2F%22%7D)

The analysis was carried out in Python. In accordance with the course instructions, code is not included in the main report but was used to generate all tables and figures.

## 8 Conclusion

In this report, we studied French and German day-ahead electricity prices in 2025 through a unified statistical framework combining linear time-series methods, extreme-value analysis, copulas, and Hawkes processes. This organization made it possible to move progressively from marginal temporal structure, to tail behavior, to cross-sectional dependence, and finally to temporal clustering of extreme events.

Our results suggest that the French and German markets are linked by more than average co-movement. In particular, dependence becomes especially informative during stress episodes, where tail behavior and event clustering matter more than standard correlation. The copula analysis highlights asymmetric dependence patterns, while the Hawkes analysis shows that extreme events are not isolated but tend to propagate over time and, to some extent, across markets.

More broadly, the report illustrates the value of combining several tools from the course around a single coherent empirical question. Rather than presenting a juxtaposition of methods, we show that each technique answers a distinct part of the same problem: temporal dynamics, extreme risk, dependence structure, and self-excitation. This makes the French-German electricity market a natural and informative case study for statistical analysis in 2025.

## Acknowledgments

We thank Mathieu Rosenbaum and the instructors of the course for their lectures and guidance. We also thank École Polytechnique for providing the academic environment in which this project was developed.

## Author Contributions

**Maxime Puech** conducted the linear time-series analysis, including the study of hourly and weekly seasonality, autocorrelation structure, and the preparation of transformed series used in later sections.

**Thaïs Roussel** carried out the extreme-value analysis, including the study of negative prices, large positive spikes, threshold exceedances, and the statistical characterization of rare events used to motivate the event definitions in the Hawkes section.

**Lucas Jung** conducted the copula-based dependence analysis and the Hawkes-process analysis, including the construction of pseudo-observations, the estimation and comparison of Gaussian, Gumbel, and Clayton copulas, and the modeling of temporal clustering and cross-excitation of extreme events.

All authors contributed to the interpretation of results, discussed the overall empirical strategy, and reviewed the final manuscript.

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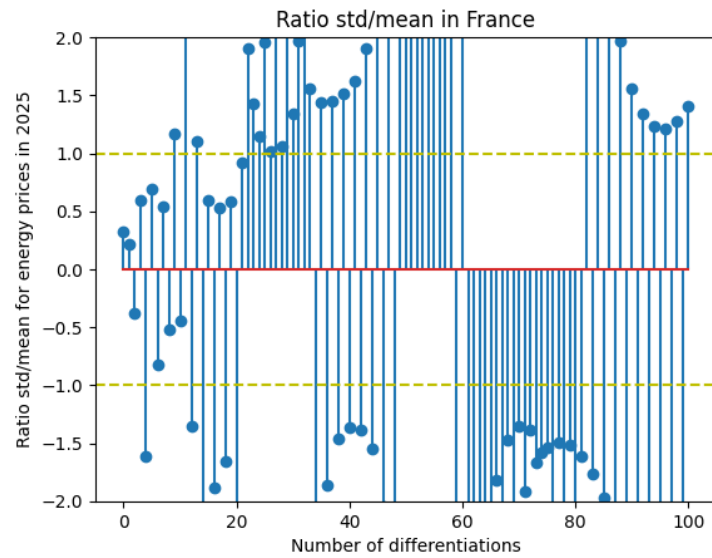


Figure 1: Ratio std/mean of the expectancy function for the french prices process differentiated  $n$  times.

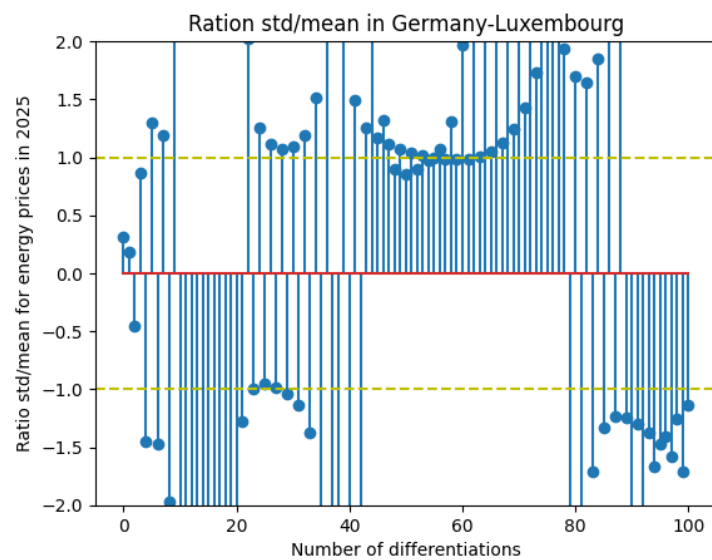


Figure 2: Ratio std/mean of the expectancy function for the german prices process differentiated  $n$  times.



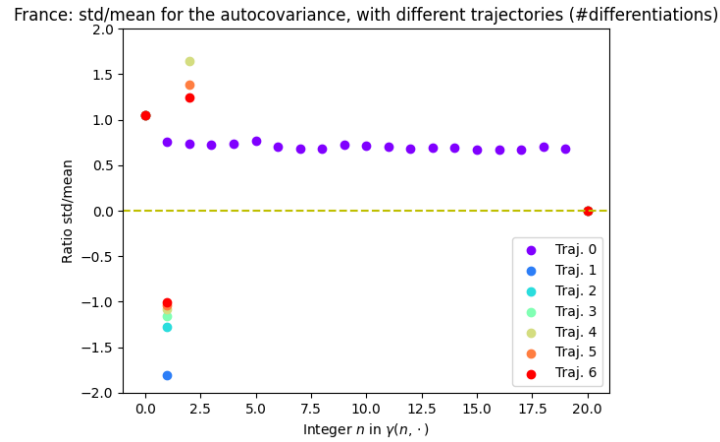


Figure 3: Ratio std/mean of the autocovariance function for the french prices process, with trajectories depending on the number of differentiations.

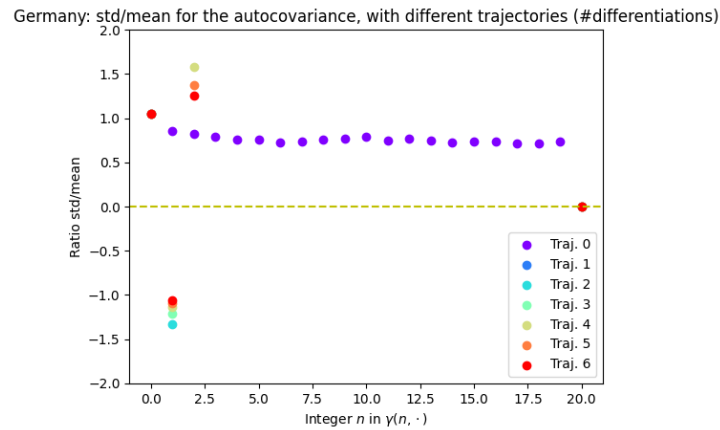


Figure 4: Ratio std/mean of the autocovariance function for the german prices process, with trajectories depending on the number of differentiations.

```

Best (p,q): (5, 5)

SARIMAX Results
=====
Dep. Variable:          y      No. Observations:      35040
Model:                ARIMA(5, 0, 5)  Log Likelihood      -125540.004
Date:                Thu, 26 Mar 2026  AIC              251104.009
Time:                18:05:39    BIC              251205.580
Sample:              0      HQIC              251136.360
                    - 35040
Covariance Type:      opg
=====
              coef    std err          z      P>|z|      [0.025    0.975]
-----
const         61.0808      2.060      29.652      0.000      57.043     65.118
ar.L1          0.8730      0.047      18.696      0.000       0.781     0.965
ar.L2          0.1655      0.043       3.845      0.000       0.081     0.250
ar.L3         -0.0319      0.018      -1.783      0.075      -0.067     0.003
ar.L4          0.2612      0.014      18.189      0.000       0.233     0.289
ar.L5         -0.2930      0.008     -38.179      0.000      -0.308    -0.278
ma.L1         -0.0036      0.047      -0.078      0.938      -0.095     0.088
ma.L2         -0.0592      0.013      -4.471      0.000      -0.085    -0.033
ma.L3         -0.0476      0.014      -3.518      0.000      -0.074    -0.021
ma.L4          0.0942      0.007      12.671      0.000       0.080     0.109
ma.L5          0.0422      0.018       2.319      0.020       0.007     0.078
sigma2        75.7550      0.157     483.554      0.000      75.448     76.062
...
=====

```

Figure 5: Results for the ARMA function of the statsmodels module (french prices).

```

Best (p,q): (5, 5)

SARIMAX Results
=====
Dep. Variable:          y      No. Observations:      35040
Model:                ARIMA(5, 0, 5)  Log Likelihood      -150842.044
Date:                Thu, 26 Mar 2026  AIC              301708.089
Time:                19:12:02    BIC              301809.660
Sample:              0      HQIC              301740.440
                    - 35040
Covariance Type:      opg
=====
              coef    std err          z      P>|z|      [0.025    0.975]
-----
const         89.3911      1.851      48.294      0.000      85.763     93.019
ar.L1          0.9868      0.011      88.610      0.000       0.965     1.009
ar.L2         -0.1027      0.008     -12.781      0.000      -0.118    -0.087
ar.L3          0.0320      0.005       6.040      0.000       0.022     0.042
ar.L4          0.5536      0.005     105.243      0.000       0.543     0.564
ar.L5         -0.5088      0.005     -96.669      0.000      -0.519    -0.498
ma.L1         -0.4613      0.011     -41.428      0.000      -0.483    -0.440
ma.L2         -0.0271      0.005      -5.156      0.000      -0.037    -0.017
ma.L3          0.0862      0.004      20.929      0.000       0.078     0.094
ma.L4          0.0184      0.005       3.945      0.000       0.009     0.028
ma.L5          0.0276      0.006       4.767      0.000       0.016     0.039
sigma2       321.1067      0.490     655.035      0.000     320.146     322.067
...
=====

```

Figure 6: Results for the ARMA function of the statsmodels module (german prices).

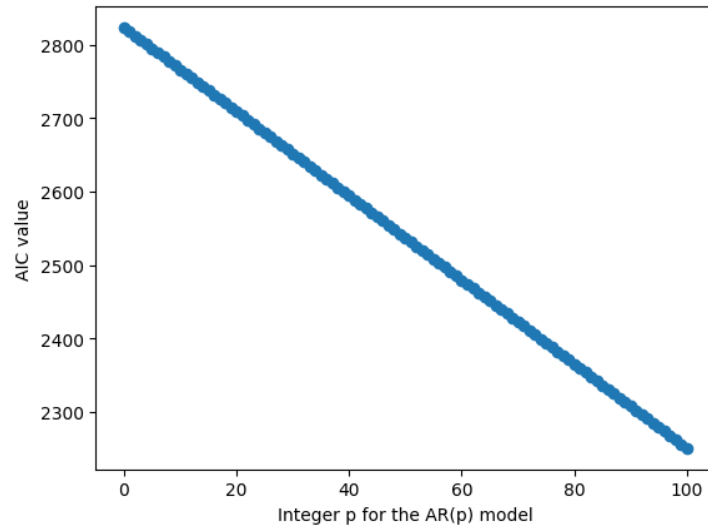


Figure 7: Results for the AIC function for french prices.

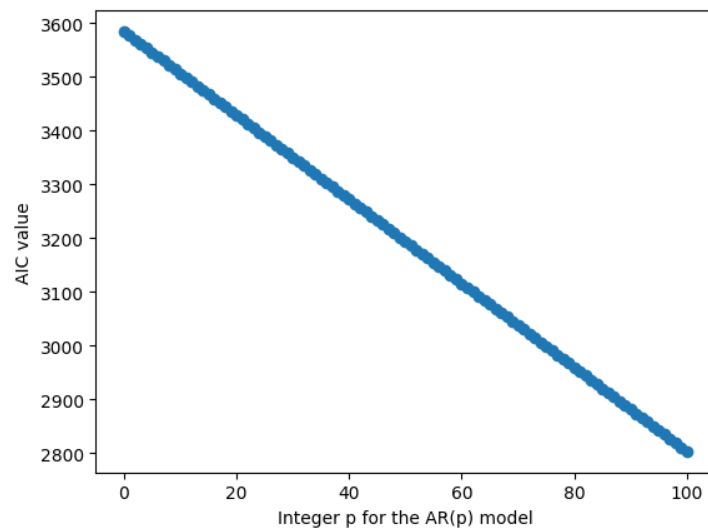


Figure 8: Results for the AIC function for german prices.

```

Best p: 25

SARIMAX Results
=====
Dep. Variable:      y      No. Observations:      35040
Model:              ARIMA(25, 0, 0)      Log Likelihood      -124987.269
Date:              Fri, 27 Mar 2026      AIC      250028.538
Time:              16:14:01      BIC      250257.072
Sample:            0      HQIC      250101.328
                  - 35040
Covariance Type:    opg
=====
              coef      std err      z      P>|z|      [0.025      0.975]
-----
const          61.0808      2.945      20.744      0.000      55.310      66.852
ar.L1           0.8703      0.003      312.639      0.000      0.865      0.876
ar.L2           0.1091      0.004      25.668      0.000      0.101      0.117
ar.L3          -0.0278      0.005      -5.573      0.000      -0.038      -0.018
ar.L4           0.3815      0.004      85.170      0.000      0.373      0.390
ar.L5          -0.3079      0.005     -68.009      0.000      -0.317      -0.299
ar.L6          -0.0240      0.006      -4.108      0.000      -0.036      -0.013
ar.L7          -0.0103      0.006      -1.643      0.100      -0.023      0.002
ar.L8          -0.0412      0.005     -7.605      0.000      -0.052      -0.031
ar.L9           0.0132      0.006      2.192      0.028      0.001      0.025
ar.L10          0.0210      0.007      2.908      0.004      0.007      0.035
ar.L11         -0.0050      0.007     -0.676      0.499      -0.019      0.009
...
=====

```

Figure 9: Results for the AR modelisation with the statsmodels module for french prices.

```

Best p: 25

SARIMAX Results
=====
Dep. Variable:      y      No. Observations:      35040
Model:              ARIMA(25, 0, 0)      Log Likelihood      -150381.666
Date:              Fri, 27 Mar 2026      AIC      300817.333
Time:              16:39:04      BIC      301045.867
Sample:            0      HQIC      300890.123
                  - 35040
Covariance Type:    opg
=====
              coef      std err      z      P>|z|      [0.025      0.975]
-----
const          89.3920      2.100      42.576      0.000      85.277      93.507
ar.L1           0.5282      0.001      407.499      0.000      0.526      0.531
ar.L2           0.1246      0.002      61.532      0.000      0.121      0.129
ar.L3           0.1747      0.002      102.225      0.000      0.171      0.178
ar.L4           0.5955      0.001      486.481      0.000      0.593      0.598
ar.L5          -0.1942      0.002     -80.193      0.000      -0.199      -0.189
ar.L6          -0.1284      0.003     -42.556      0.000      -0.134      -0.122
ar.L7          -0.1191      0.002     -49.611      0.000      -0.124      -0.114
ar.L8          -0.0216      0.003     -7.210      0.000      -0.028      -0.016
ar.L9          -0.0630      0.004     -17.798      0.000      -0.070      -0.056
ar.L10          0.0049      0.004      1.156      0.248      -0.003      0.013
ar.L11          0.0320      0.004      7.239      0.000      0.023      0.041
...
=====

```

Figure 10: Results for the AR modelisation with the statsmodels module for french prices.

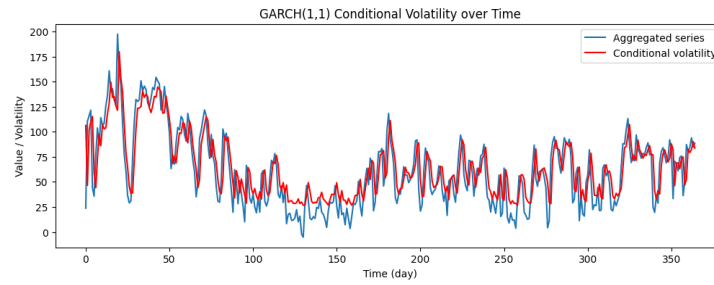


Figure 11: GARCH(1,1) modelisation for french prices.

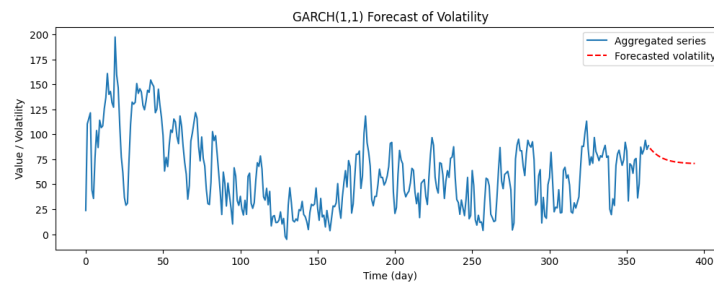


Figure 12: GARCH(1,1) forecast for french prices.

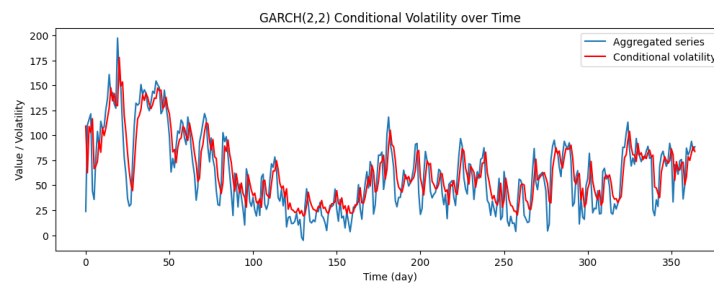


Figure 13: GARCH(2,2) modelisation for french prices.

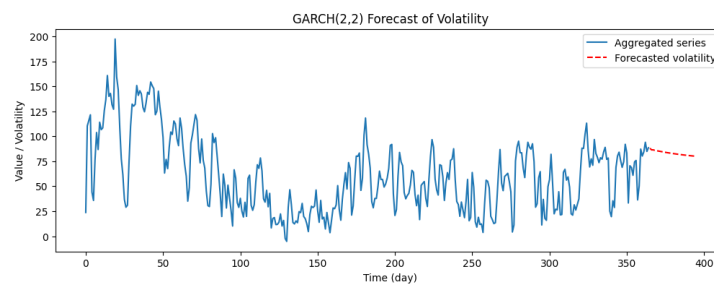


Figure 14: GARCH(2,2) forecast for french prices.

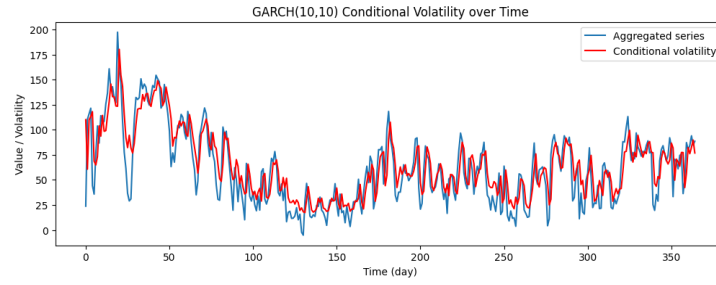


Figure 15: GARCH(10,10) modelisation for french prices.

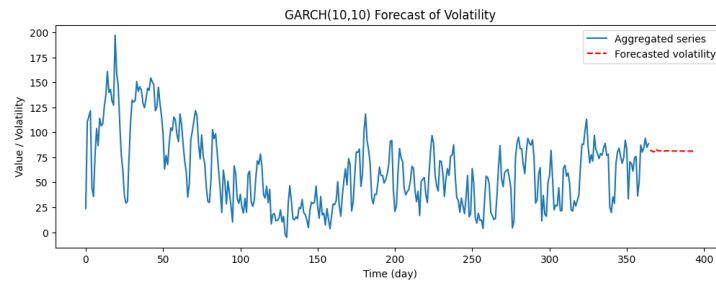


Figure 16: GARCH(10,10) forecast for french prices.

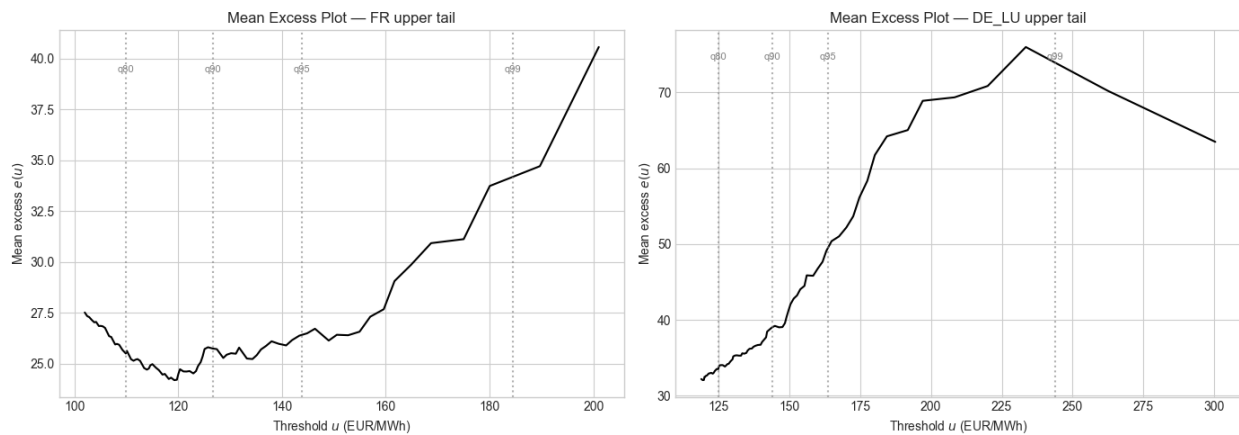


Figure 17: Mean Excess Plot for upper-tail exceedances in France and Germany. The approximately linear region identifies the range of viable thresholds.

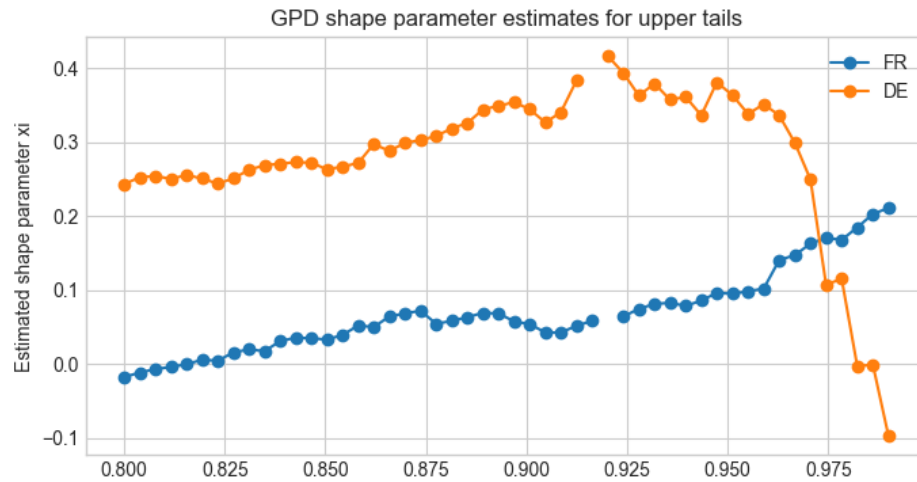


Figure 18: Estimated GPD shape parameter  $\hat{\xi}$  as a function of the quantile threshold for the upper tail. The selected threshold  $q = 0.93$  lies in a region of relative stability.

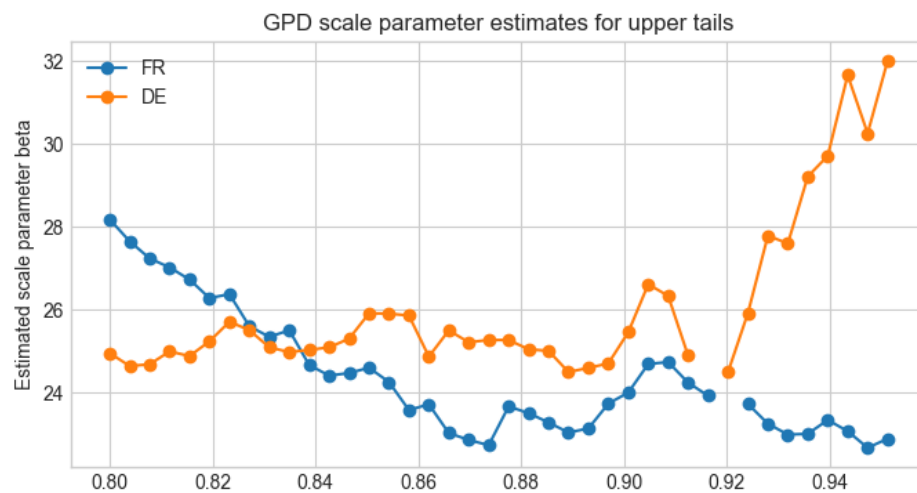


Figure 19: Estimated GPD scale parameter  $\hat{\beta}$  as a function of the quantile threshold for the upper tail.

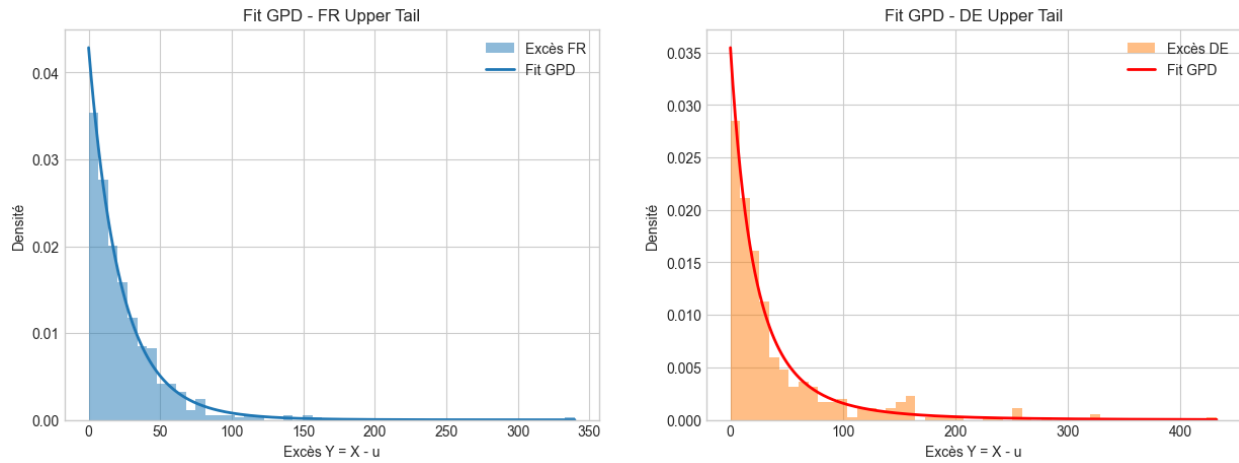


Figure 20: Histogram of upper-tail exceedances and fitted GPD density for France (left) and Germany (right) at quantile threshold  $q = 0.93$ .

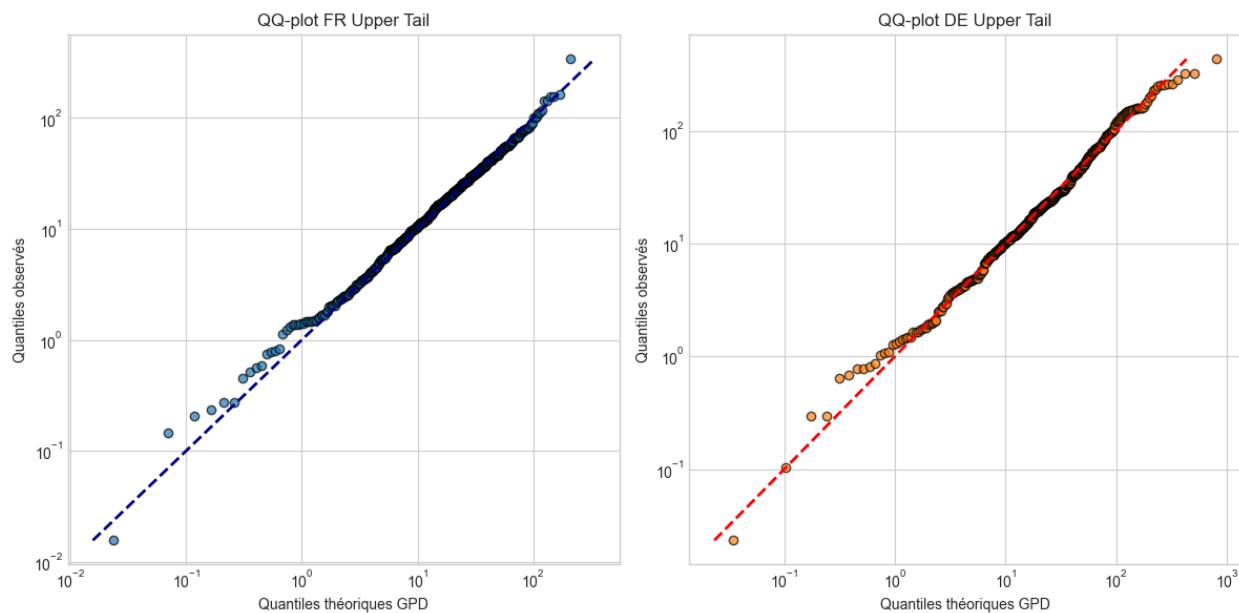


Figure 21: QQ-plot of the upper-tail exceedances against the fitted GPD quantiles for France (left) and Germany (right).



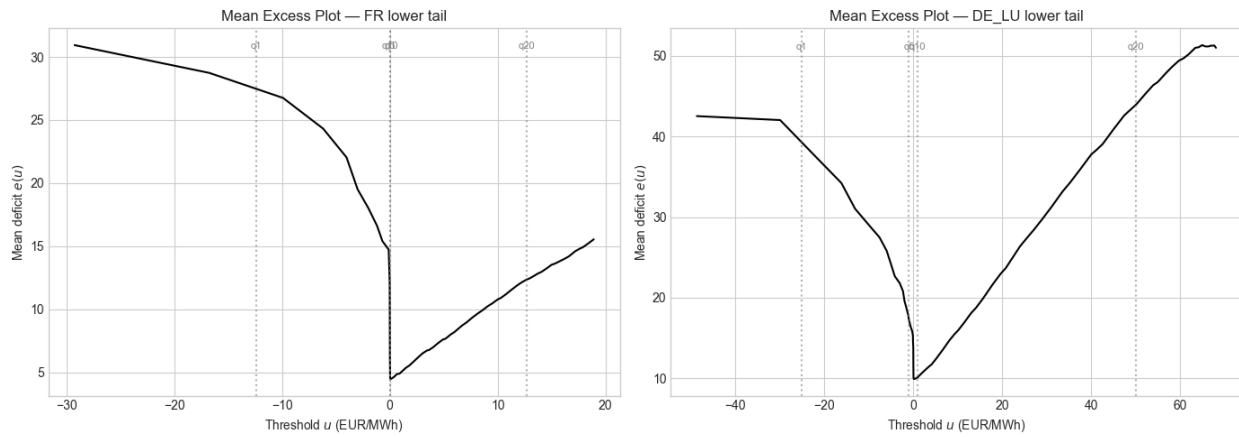


Figure 22: Mean Deficit Plot for lower-tail exceedances in France and Germany. The approximately linear region guides the threshold choice.

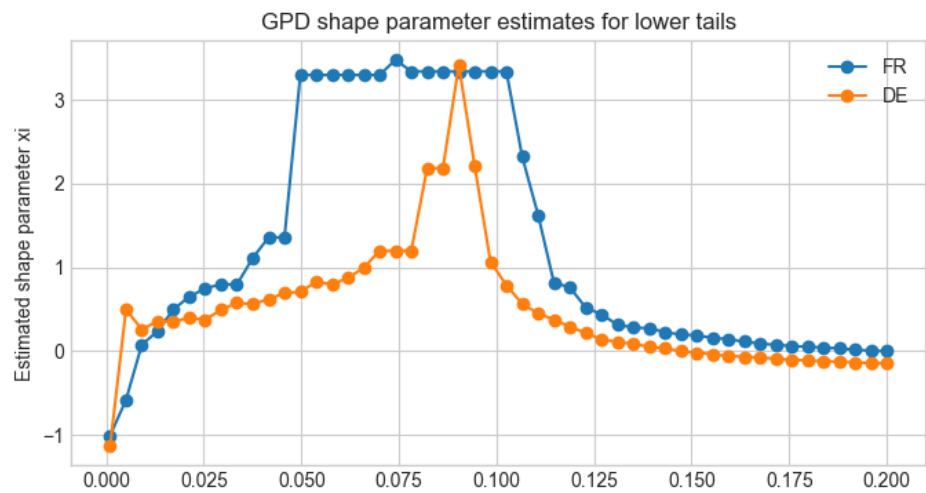


Figure 23: Estimated GPD shape parameter  $\hat{\xi}$  as a function of the quantile threshold for the lower tail.

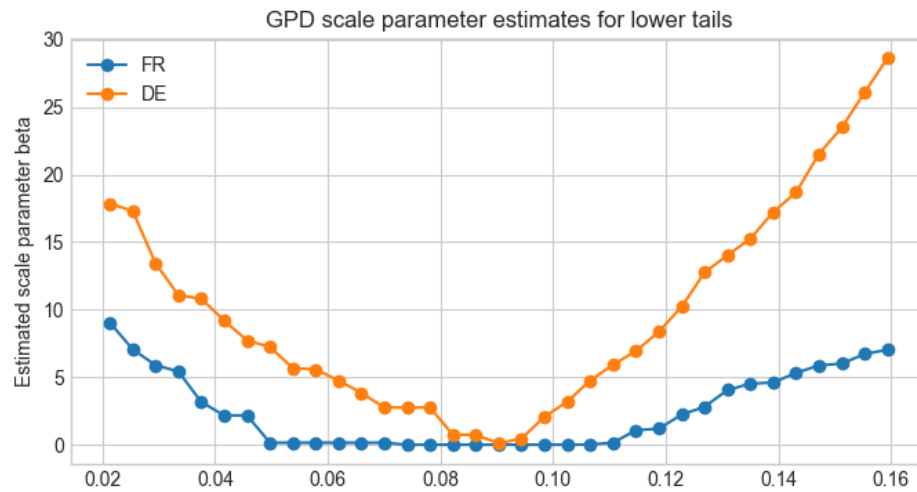


Figure 24: Estimated GPD scale parameter  $\hat{\beta}$  as a function of the quantile threshold for the lower tail.

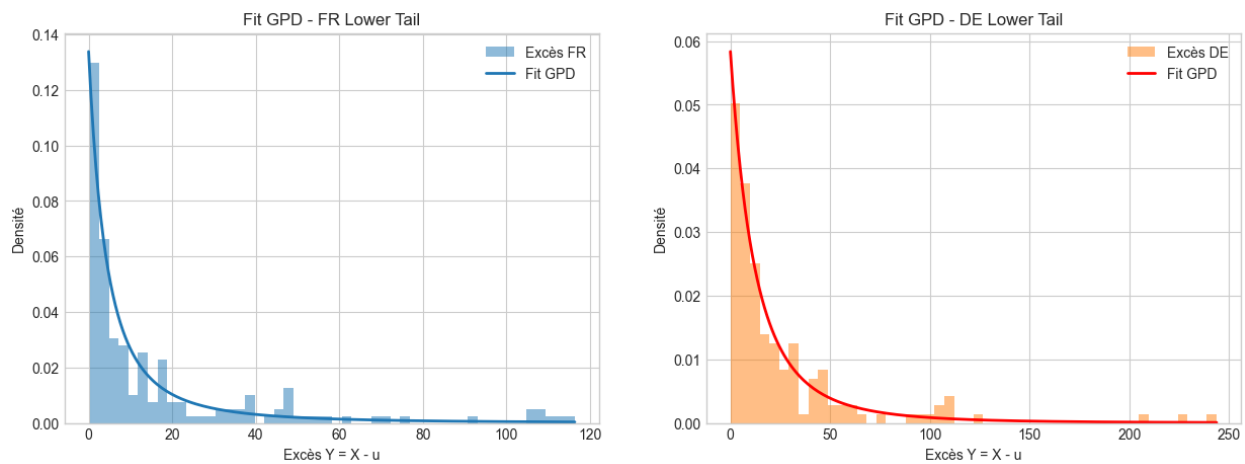


Figure 25: Histogram of lower-tail deficits and fitted GPD density for France (left) and Germany (right) at quantile threshold  $q = 0.025$ .

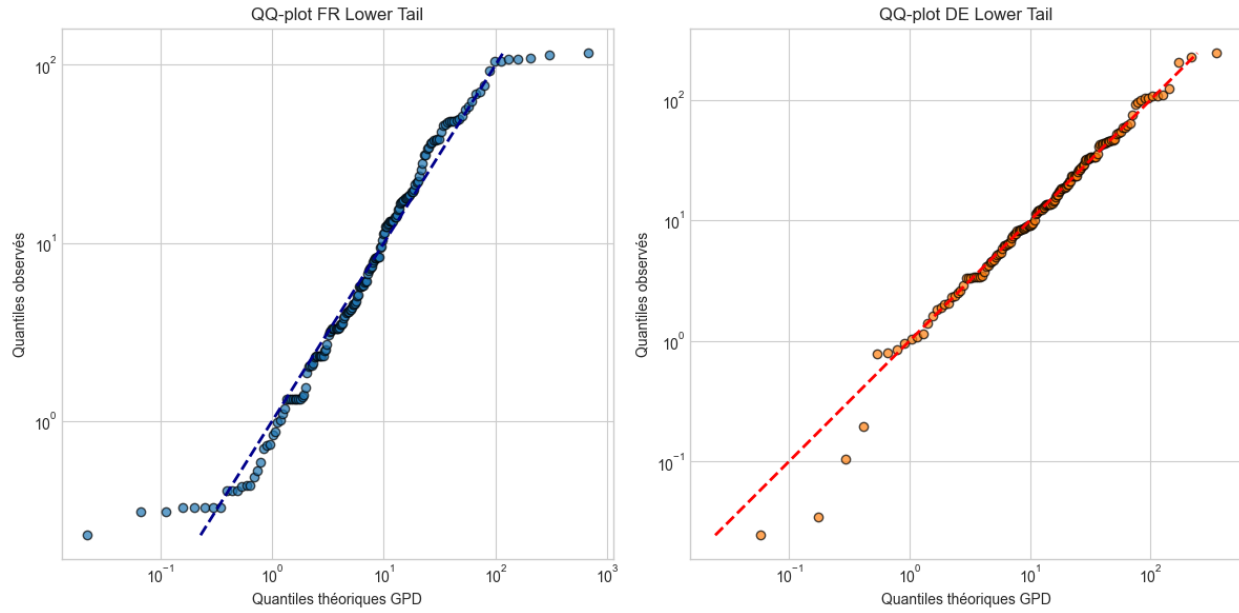


Figure 26: QQ-plot of the lower-tail deficits against the fitted GPD quantiles for France (left) and Germany (right).

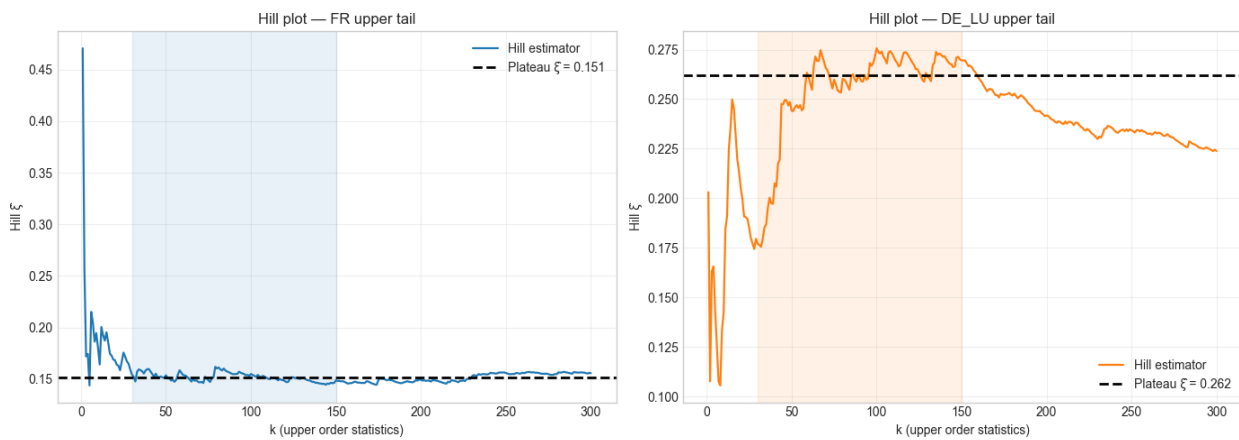


Figure 27: Hill plots for the upper tail of French and German prices. The stable plateau in the range  $k \in [30, 150]$  provides the Hill estimate.

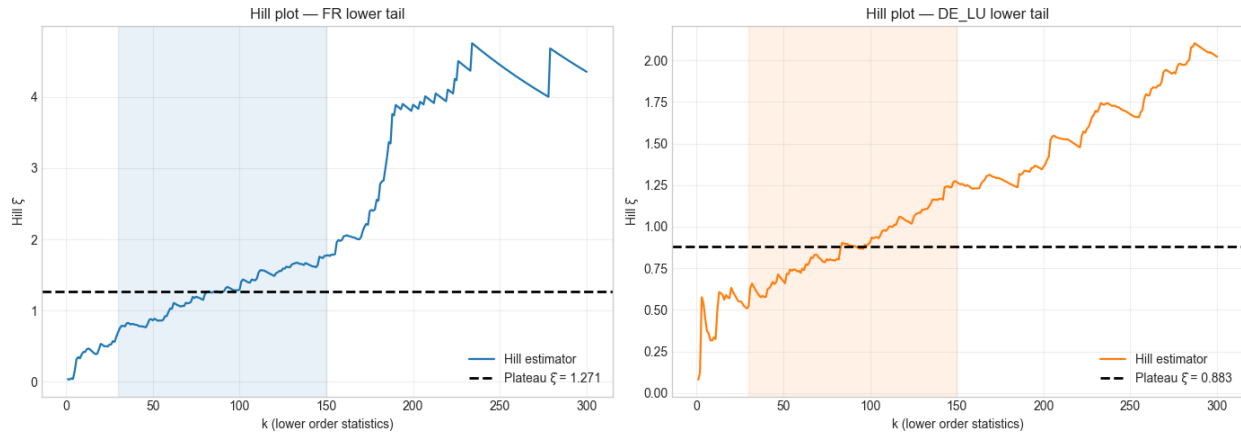


Figure 28: Hill plots for the lower tail (reflected series) of French and German prices.

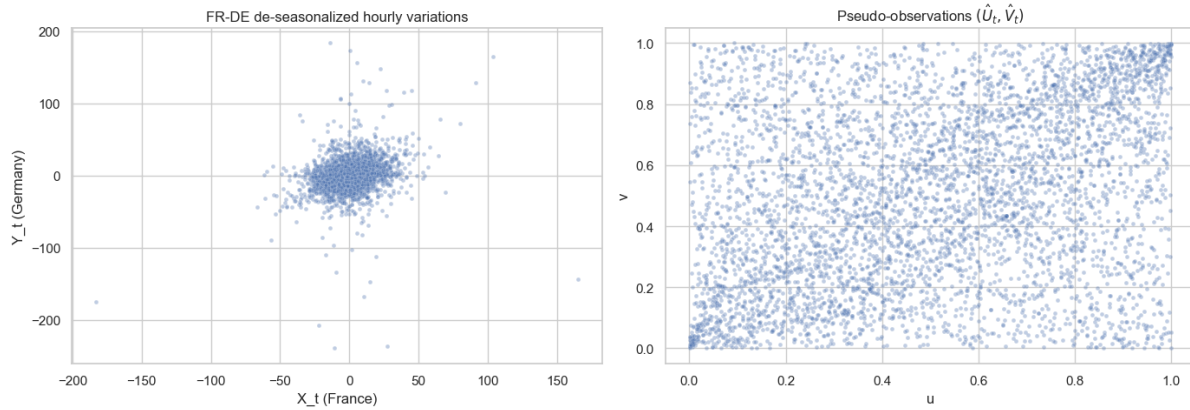


Figure 29: Left: scatter plot of de-seasonalized hourly price variations in France and Germany. Right: pseudo-observations  $(\hat{U}_t, \hat{V}_t)$  used for copula estimation.

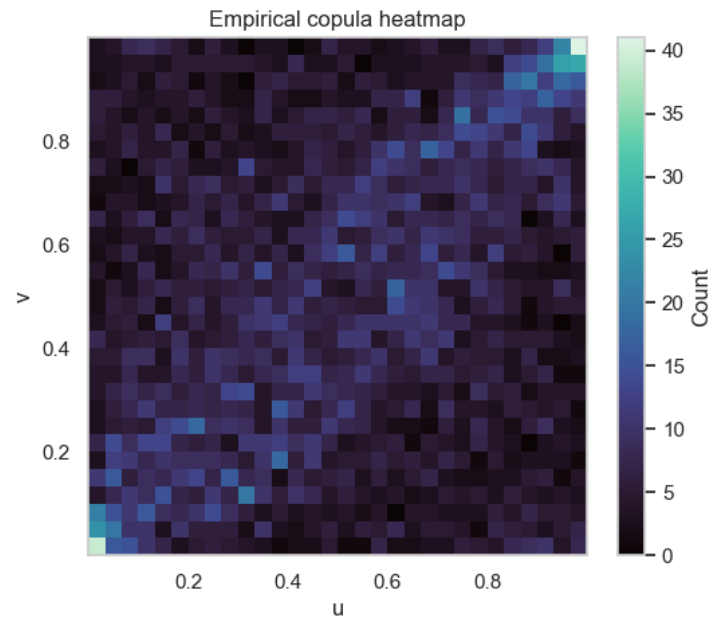


Figure 30: Empirical copula heatmap based on the pseudo-observations.

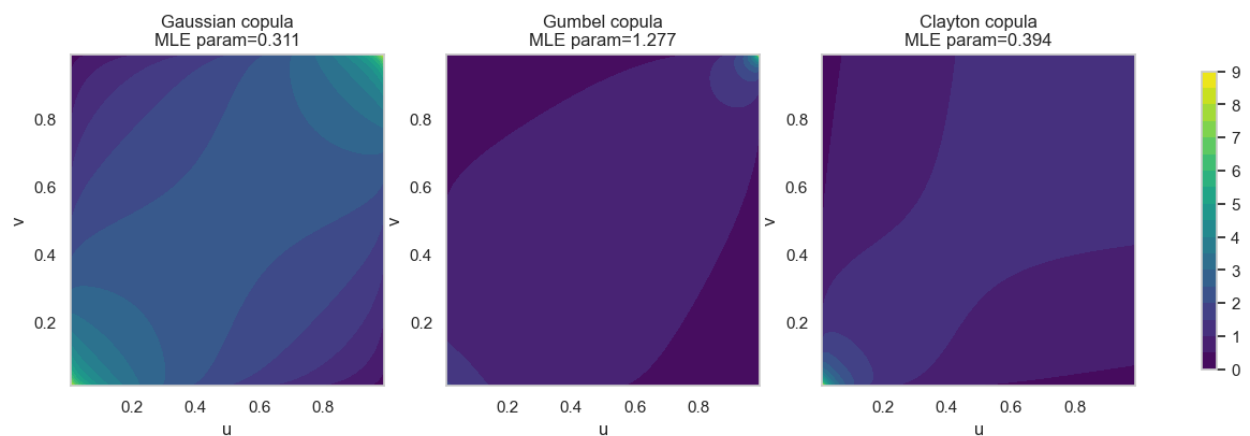


Figure 31: Fitted copula contours for the Gaussian, Gumbel, and Clayton models.

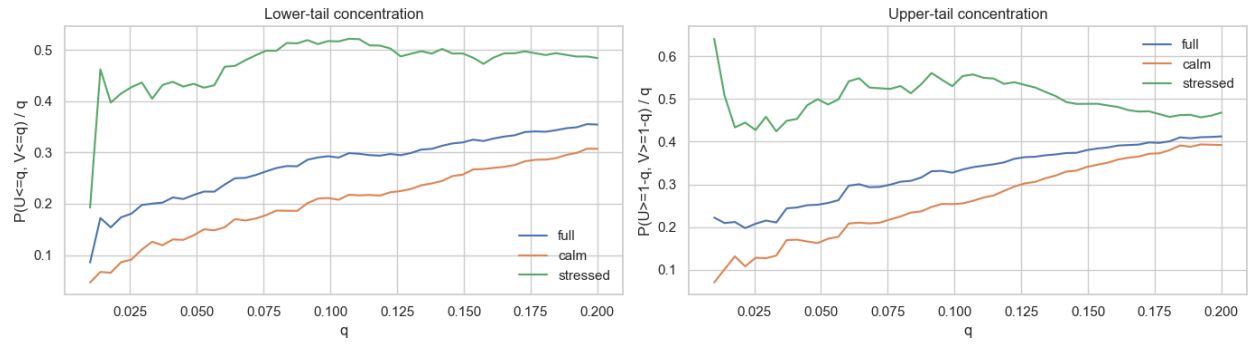


Figure 32: Empirical lower-tail and upper-tail concentration functions across regimes.

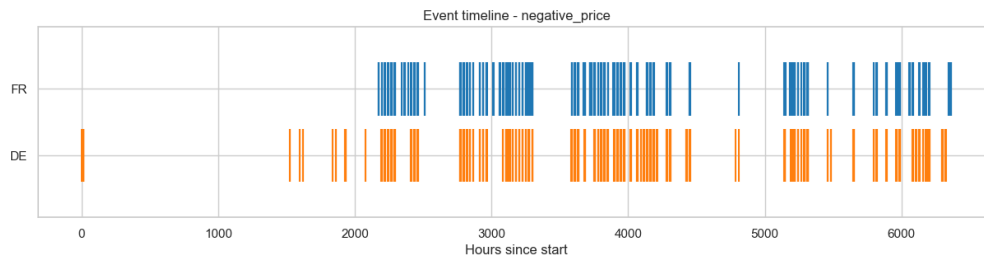


Figure 33: Timeline of negative-price events in France and Germany over the full sample.

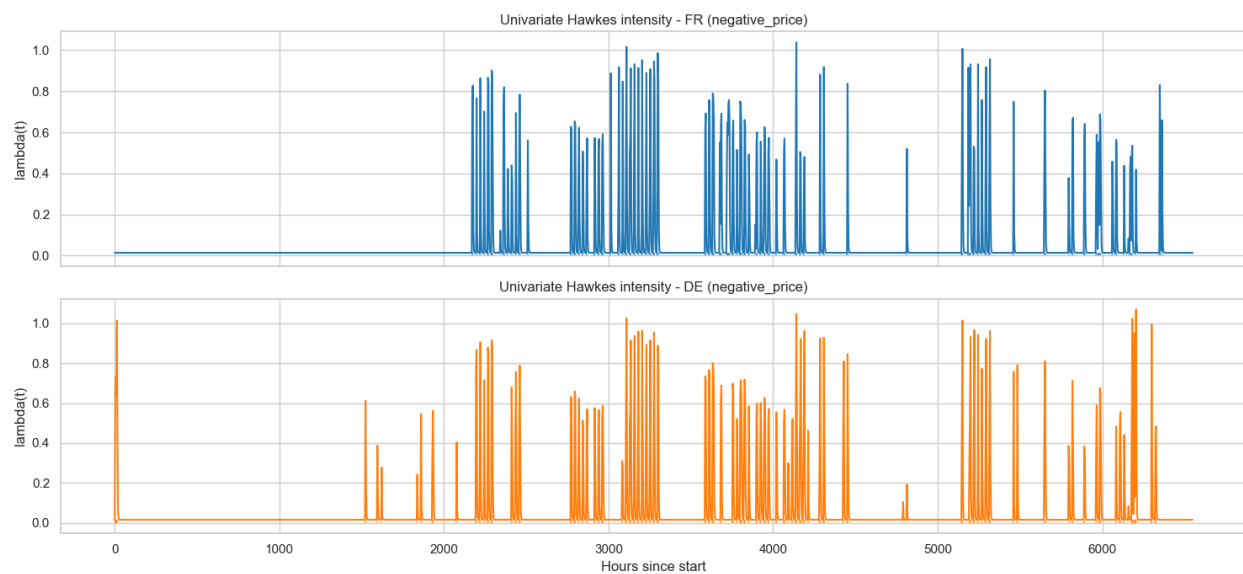


Figure 34: Estimated univariate Hawkes intensities for negative-price events in France and Germany.

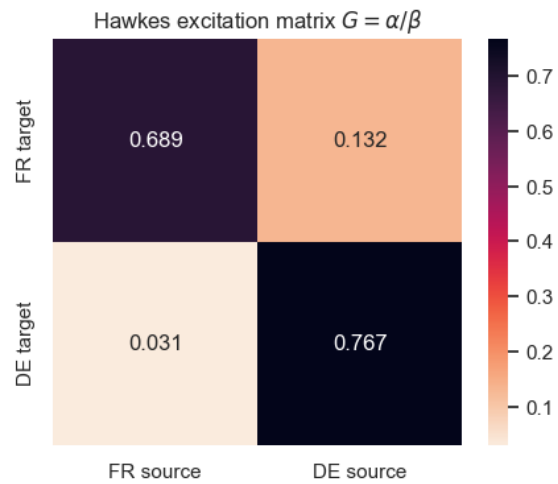


Figure 35: Estimated excitation matrix for the bivariate Hawkes model on negative-price events.

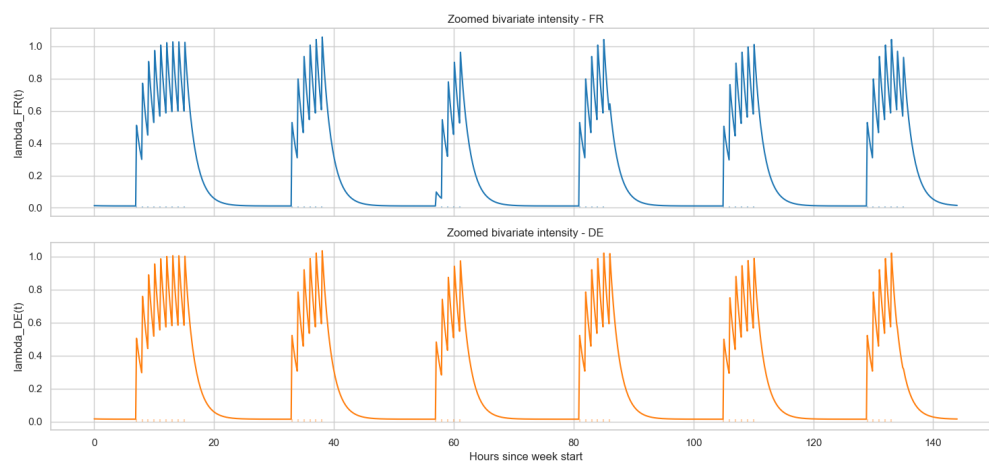


Figure 36: Zoom on a tense week for negative-price events and fitted bivariate intensities.